ASDD PRACTICAL EXAM EXPERIMENT LIST

For Github Experiments: Check whether you have forked the given repository: https://github.com/shruthi1722/node_test.git

Otherwise Fork and Clone the repository in your local machine during practical exam.

1. Jira Board and Sprint Management:

Create a Jira board for a given project, create Product and Sprint Backlogs, add 5-6 user stories, start a sprint, move stories through To Do, In Progress, In Review, and Done, and add appropriate comments.

Theory Question: Differentiate between Product Backlog and Sprint Backlog. How is the Sprint Process carried out in an Agile Team?

Theory Question: Define Sprint Backlog and Product Backlog?

2. Jira and GitHub Integration:

Create a Jira Story and add it to To Do. Clone the forked repository (mentioned repository url above), create a github branch using Jira story. Follow the Github Integration steps.

Theory Question: How can Jira issues be linked to GitHub branches and commits? How it is useful for others to track the changes.

3. Git Commands and Version Control:

Clone the forked repository (mentioned repository url above). Use Github commands to create a github branch, list branches.

Create or modify file and use github commands to check status, add the file in

the staging environment and commit changes, and push the branch to GitHub.

Commands:

git checkout -b branch_name

git branch

git add .

git commit -m “”

git push

Theory Question: Explain the purpose of git add, git commit, git push, git branch, and git status.

Theory Question : Explain Github Actions

4. Docker Image Management and Container Execution:

Pull a Docker image, list the available Docker images, display the running containers, run a container using the downloaded image, and verify that the container is running successfully.

Commands:

docker pull

docker images

docker run -it bash image_name

docker ps

Theory Question: Differentiate between a Docker image and a Docker container.

Theory Question:Explain Docker Build command

5. Dockerfile Creation and Image Building:

Create a Dockerfile for a given application, build a Docker image, run a container using the image, and verify that the application is working.

Theory Question: What is a Dockerfile, and explain the purpose of the FROM, COPY, RUN, EXPOSE, and CMD instructions?

.6. Terraform and GitHub Actions:

Fork the Terraform project https://github.com/shruthi1722/terraform23, clone it to the local machine, make changes to the main.tf file, commit and push the changes, and verify the GitHub Actions workflow execution.

Theory Question: What is Terraform? List the Terraform commands and explain their purpose briefly.

7. Grafana Using Docker:

Pull and run the Grafana Docker image, map the container port to localhost:3001, and verify that the Grafana dashboard is accessible in a web browser.

Theory Question: What is Grafana, and how is it used for monitoring and visualization in DevOps?

Before starting with any github related experiments , do the following

Steps to Generate a GitHub Token for git push

1. Open https://github.com/settings/tokens
2. Sign in to your GitHub account.
3. Click Tokens (classic).
4. Click Generate new token → Generate new token (classic).
5. Enter a note, such as Git Push Token.
6. Select the token expiration period.
7. Under Select scopes, tick the repo checkbox.
8. Scroll down and click Generate token.
9. Copy the generated token immediately. You will not be able to view it again.

Steps to Use the Token

10. Open your terminal and navigate to your Git repository.
11. Run:

git push

12. When prompted, enter your GitHub username.
13. When prompted for a password, paste the generated token instead of your GitHub password.
14. Press Enter.